

03 Raw Data Storage

Course Name: Data Management for Machine Learning

Assignment Title: End-to-End Data Management Pipeline for a Recommendation System

Assignment: Group 51 - Data management for Machine Learning Group 51

Team Members

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1. Objective

This report documents the raw data storage layer for the recommendation pipeline. It captures evidence from the project structure and available logs to show how ingested data is stored, how the folder layout is organized, and which scripts or configuration files support the storage process.

2. Raw Data Storage Objective Coverage

- Store ingested data in a local data lake or cloud-like local filesystem structure.
- Use a structured folder layout partitioned by source, type, and timestamp where available.
- Provide storage structure documentation.
- Provide upload scripts or configuration file evidence.

3. Raw Data Storage Summary

- Raw files discovered: 6
- Sources identified: dummyjson, retailrocket
- Timestamp partition load_date found: Yes
- Timestamp partition load_hour found: Yes
- File types found: .csv, .json

4. Raw Data Files Identified

The table below lists discovered raw data assets under the local raw storage layer. Long file names and paths are wrapped explicitly so they remain readable in the PDF.

Source	File Name	Relative Path	Type	Last Modified
dummyjson	categories_raw.json	data\raw\dummyjson\load_date=2026-04-29\load_hour=11\categories_raw.json	.json	2026-04-29 16:42:35
dummyjson	products_raw.json	data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json	.json	2026-04-29 16:42:35

Source	File Name	Relative Path	Type	Last Modified
retailrocket	category_tree.csv	data\raw\retailrocket\load_date=2026-04-29\load_hour=11\category_tree.csv	.csv	2026-04-20 17:27:59
retailrocket	events.csv	data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv	.csv	2026-04-20 17:28:00
retailrocket	item_properties_part1.csv	data\raw\retailrocket\load_date=2026-04-29\load_hour=11\item_properties_part1.csv	.csv	2026-04-20 17:28:03
retailrocket	item_properties_part2.csv	data\raw\retailrocket\load_date=2026-04-29\load_hour=11\item_properties_part2.csv	.csv	2026-04-20 17:28:05

5. Storage Structure Documentation

The raw storage tree below documents the folder layout used for ingested files. This provides evidence of source-based organization and timestamp-based partitions where those folders exist.

```

raw/
├── dummyjson/
│   ├── load_date=2026-04-29/
│   │   ├── load_hour=11/
│   │   │   ├── categories_raw.json
│   │   │   └── products_raw.json
│   └── retailrocket/
│       ├── load_date=2026-04-29/
│       │   ├── load_hour=11/
│       │   │   ├── category_tree.csv
│       │   │   ├── events.csv
│       │   │   ├── item_properties_part1.csv
│       │   │   └── item_properties_part2.csv

```

6. Upload Scripts or Configuration Files

The following scripts, notebooks, and configuration files were identified as likely storage, upload, ingestion, or load-related assets in the project.

Script / Config	Relative Path	Type	Last Modified	Size
categories_raw.json	data\raw\dummyjson\load_date=2026-04-29\load_hour=11\categories_raw.json	Config	2026-04-29 16:42:35	3.13 KB
products_raw.json	data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json	Config	2026-04-29 16:42:35	416.89 KB
categories_raw.json	data\remediated\raw\categories_raw.json	Config	2026-04-29 16:46:38	0.0 KB
products_raw.json	data\remediated\raw\products_raw.json	Config	2026-04-29 16:46:38	415.26 KB
environment.yml	environment.yml	Config	2026-04-29 16:09:16	0.6 KB
feature_registry.yaml	feature_store\registry\feature_registry.yaml	Config	2026-04-28 22:39:16	5.26 KB
feature_store_20260429T072423Z.json	logs\feature_store_20260429T072423Z.json	Config	2026-04-29 13:18:24	0.18 KB
ingestion_20260429T072423Z.json	logs\ingestion_20260429T072423Z.json	Config	2026-04-29 13:14:19	1.07 KB
model_training_20260429T072423Z.json	logs\model_training_20260429T072423Z.json	Config	2026-04-29 13:18:24	0.14 KB
compare_models_started_20260429_164131.json	logs\monitoring\compare_models_started_20260429_164131.json	Config	2026-04-29 17:07:04	0.4 KB

Script / Config	Relative Path	Type	Last Modified	Size
compare_models_success_20260429_164131.json	logs\monitoring\compare_models_success_20260429_164131.json	Config	2026-04-29 17:07:14	0.4 KB
eda_prep_failed_20260429_115432.json	logs\monitoring\eda_prep_failed_20260429_115432.json	Config	2026-04-29 12:03:28	3.76 KB
eda_prep_failed_20260429_121758.json	logs\monitoring\eda_prep_failed_20260429_121758.json	Config	2026-04-29 12:23:35	3.76 KB
eda_prep_failed_20260429_132217.json	logs\monitoring\eda_prep_failed_20260429_132217.json	Config	2026-04-29 13:28:13	3.72 KB
eda_prep_started_20260429_115432.json	logs\monitoring\eda_prep_started_20260429_115432.json	Config	2026-04-29 12:02:55	0.39 KB
eda_prep_started_20260429_121758.json	logs\monitoring\eda_prep_started_20260429_121758.json	Config	2026-04-29 12:23:26	0.39 KB
eda_prep_started_20260429_132217.json	logs\monitoring\eda_prep_started_20260429_132217.json	Config	2026-04-29 13:27:49	0.39 KB
eda_prep_started_20260429_160933.json	logs\monitoring\eda_prep_started_20260429_160933.json	Config	2026-04-29 16:14:03	0.39 KB
eda_prep_started_20260429_164131.json	logs\monitoring\eda_prep_started_20260429_164131.json	Config	2026-04-29 16:47:46	0.39 KB
eda_prep_success_20260429_160933.json	logs\monitoring\eda_prep_success_20260429_160933.json	Config	2026-04-29 16:16:51	0.39 KB
eda_prep_success_20260429_164131.json	logs\monitoring\eda_prep_success_20260429_164131.json	Config	2026-04-29 16:51:47	0.39 KB
ingest_dummyjson_started_20260429_115432.json	logs\monitoring\ingest_dummyjson_started_20260429_115432.json	Config	2026-04-29 11:55:29	0.42 KB
ingest_dummyjson_started_20260429_121758.json	logs\monitoring\ingest_dummyjson_started_20260429_121758.json	Config	2026-04-29 12:18:03	0.42 KB
ingest_dummyjson_started_20260429_132217.json	logs\monitoring\ingest_dummyjson_started_20260429_132217.json	Config	2026-04-29 13:22:20	0.42 KB
ingest_dummyjson_started_20260429_155743.json	logs\monitoring\ingest_dummyjson_started_20260429_155743.json	Config	2026-04-29 15:58:49	0.42 KB

Preview: categories_raw.json

Path: data\raw\dummyjson\load_date=2026-04-29\load_hour=11\categories_raw.json

```
[
  {
    "slug": "beauty",
    "name": "Beauty",
    "url": "https://dummyjson.com/products/category/beauty"
  },
  {
    "slug": "fragrances",
    "name": "Fragrances",
    "url": "https://dummyjson.com/products/category/fragrances"
  },
  {
    "slug": "furniture",
    "name": "Furniture",
    "url": "https://dummyjson.com/products/category/furniture"
  },
  {
    "slug": "groceries",
    "name": "Groceries",
    "url": "https://dummyjson.com/products/category/groceries"
  },
  {
    "slug": "home-decoration",
    "name": "Home Decoration",
    "url": "https://dummyjson.com/products/category/home-decoration"
  },
]
```

```
{
  "slug": "kitchen-accessories",
  "name": "Kitchen Accessories",
  "url": "https://dummyjson.com/products/category/kitchen-accessories"
},
{
  "slug": "laptops",
  "name": "Laptops",
  "url": "https://dummyjson.com/products/category/laptops"
},
{
  "slug": "mens-shirts",
  "name": "Mens Shirts",
  "url": "https://dummyjson.com/products/category/mens-shirts"
```

Preview: products_raw.json

Path: data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json

```
{
  "products": [
    {
      "id": 1,
      "title": "Essence Mascara Lash Princess",
      "description": "The Essence Mascara Lash Princess is a popular mascara known for its volumizing and lengthening effects. Achieve dramatic lashes with this long-lasting and cruelty-free formula.",
      "category": "beauty",
      "price": 9.99,
      "discountPercentage": 10.48,
      "rating": 2.56,
      "stock": 99,
      "tags": [
        "beauty",
        "mascara"
      ],
      "brand": "Essence",
      "sku": "BEA-ESS-ESS-001",
      "weight": 4,
      "dimensions": {
        "width": 15.14,
        "height": 13.08,
        "depth": 22.99
      },
      "warrantyInformation": "1 week warranty",
      "shippingInformation": "Ships in 3-5 business days",
      "availabilityStatus": "In Stock",
      "reviews": [
        {
          "rating": 3,
          "comment": "Would not recommend!",
          "date": "2025-04-30T09:41:02.053Z",
          "reviewerName": "Eleanor Collins",
          "reviewerEmail": "eleanor.collins@x.dummyjson.com"
        },
        {
          "rating": 4,
          "comment": "Very satisfied!",
          "date": "2025-04-30T09:41:02.053Z",
          "reviewerName": "Lucas Gordon",
          "reviewerEmail": "lucas.gordon@x.dummyjson.com"
        }
      ]
    }
  ]
}
```

Preview: categories_raw.json

Path: data\remediated\raw\categories_raw.json

```
[]
```

Preview: products_raw.json

Path: data\remediated\raw\products_raw.json

```
{
  "products": [
    {
      "id": 1,
      "title": "Essence Mascara Lash Princess",
      "description": "The Essence Mascara Lash Princess is a popular mascara known for its volumizing and lengthening effects. Achieve dramatic lashes with this long-lasting and cruelty-free formula.",
      "category": "beauty",
      "price": 9.99,
      "discountPercentage": 10.48,
      "rating": 2.56,
```

```

"stock": 99,
"tags": [
  "beauty",
  "mascara"
],
"brand": "Essence",
"sku": "BEA-ESS-ESS-001",
"weight": 4,
"warrantyInformation": "1 week warranty",
"shippingInformation": "Ships in 3-5 business days",
"availabilityStatus": "In Stock",
"reviews": [
  {
    "rating": 3,
    "comment": "Would not recommend!",
    "date": "2025-04-30T09:41:02.053Z",
    "reviewerName": "Eleanor Collins",
    "reviewerEmail": "eleanor.collins@x.dummyjson.com"
  },
  {
    "rating": 4,
    "comment": "Very satisfied!",
    "date": "2025-04-30T09:41:02.053Z",
    "reviewerName": "Lucas Gordon",
    "reviewerEmail": "lucas.gordon@x.dummyjson.com"
  },
  {
    "rating": 5,
    "comment": "Highly impressed!",
    "date": "2025-04-30T09:41:02.053Z",
  }
]

```

Preview: environment.yml

Path: environment.yml

```

name: dm4ml
channels:
  - defaults
dependencies:
  - python=3.11
  - pip
  - numpy
  - pandas
  - scipy
  - scikit-learn
  - matplotlib
  - seaborn
  - jupyter
  - notebook
  - ipykernel
  - pyarrow
  - reportlab
  - pip:
    - mlflow
    - prefect
    - papermill
    - kagglehub

# run these
conda env create -f environment.yml
conda activate dm4ml
python -m ipykernel install --user --name dm4ml --display-name "Python (dm4ml)"
python -m pip install mlflow
python -m pip install mlflow ipykernel
python -m ipykernel install --user --name dm4ml --display-name "Python (dm4ml)"

```

7. Logs and Reports Related to Storage

The table below lists discovered logs and generated reports from the project. Wrapped table cells are used so long report names and relative paths do not get cut off.

Log / Report File	Relative Path	Last Modified	Size
ingestion_summary.csv	data\bronze\dummyjson\ingestion_summary.csv	2026-04-29 16:42:35	0.23 KB
ingestion_summary.csv	data\bronze\retailrocket\ingestion_summary.csv	2026-04-29 16:43:31	0.33 KB
prepared_validation_issues_20260429T111817Z.csv	data\quality\prepared_validation_issues_20260429T111817Z.csv	2026-04-29 16:50:57	0.0 KB
prepared_validation_summary_20260429T111817Z.csv	data\quality\prepared_validation_summary_20260429T111817Z.csv	2026-04-29 16:50:57	0.16 KB
raw_validation_issues_20260429T111817Z.csv	data\quality\raw_validation_issues_20260429T111817Z.csv	2026-04-29 16:49:34	0.73 KB
raw_validation_summary_20260429T111817Z.csv	data\quality\raw_validation_summary_20260429T111817Z.csv	2026-04-29 16:49:34	0.21 KB
compare_models_20260429_164131.log	logs\orchestration\compare_models_20260429_164131.log	2026-04-29 17:07:14	1.01 KB
eda_prep_20260429_115432.log	logs\orchestration\eda_prep_20260429_115432.log	2026-04-29 13:28:13	30.1 KB
eda_prep_20260429_160933.log	logs\orchestration\eda_prep_20260429_160933.log	2026-04-29 16:16:51	0.98 KB
eda_prep_20260429_164131.log	logs\orchestration\eda_prep_20260429_164131.log	2026-04-29 16:51:47	0.98 KB
ingest_dummyjson_20260429_115432.log	logs\orchestration\ingest_dummyjson_20260429_115432.log	2026-04-29 13:23:10	3.13 KB
ingest_dummyjson_20260429_155743.log	logs\orchestration\ingest_dummyjson_20260429_155743.log	2026-04-29 15:59:10	1.05 KB
ingest_dummyjson_20260429_160450.log	logs\orchestration\ingest_dummyjson_20260429_160450.log	2026-04-29 16:05:49	1.05 KB
ingest_dummyjson_20260429_160933.log	logs\orchestration\ingest_dummyjson_20260429_160933.log	2026-04-29 16:10:42	1.05 KB
ingest_dummyjson_20260429_163859.log	logs\orchestration\ingest_dummyjson_20260429_163859.log	2026-04-29 16:39:35	0.42 KB
ingest_dummyjson_20260429_164131.log	logs\orchestration\ingest_dummyjson_20260429_164131.log	2026-04-29 16:42:36	1.05 KB
ingest_retailrocket_20260429_115432.log	logs\orchestration\ingest_retailrocket_20260429_115432.log	2026-04-29 13:24:18	3.24 KB
ingest_retailrocket_20260429_155743.log	logs\orchestration\ingest_retailrocket_20260429_155743.log	2026-04-29 16:00:01	10.0 KB
ingest_retailrocket_20260429_160450.log	logs\orchestration\ingest_retailrocket_20260429_160450.log	2026-04-29 16:06:58	1.09 KB
ingest_retailrocket_20260429_160933.log	logs\orchestration\ingest_retailrocket_20260429_160933.log	2026-04-29 16:11:35	1.09 KB
ingest_retailrocket_20260429_163859.log	logs\orchestration\ingest_retailrocket_20260429_163859.log	2026-04-29 16:39:35	0.44 KB
ingest_retailrocket_20260429_164131.log	logs\orchestration\ingest_retailrocket_20260429_164131.log	2026-04-29 16:43:34	1.09 KB
materialize_features_20260429_160933.log	logs\orchestration\materialize_features_20260429_160933.log	2026-04-29 16:25:43	1.1 KB
materialize_features_20260429_164131.log	logs\orchestration\materialize_features_20260429_164131.log	2026-04-29 17:05:20	1.1 KB

Log / Report File	Relative Path	Last Modified	Size
prefect_server_20260429_114449.log	logs\orchestration\prefect_server_20260429_114449.log	2026-04-29 11:46:00	0.4 KB

8. Latest Log Preview

Long log lines are manually wrapped before being sent to the PDF preformatted block so that the output remains presentable.

```

2026-04-29 17:07:04,258 | INFO | compare_models | Starting stage: compare_models
2026-04-29 17:07:04,262 | INFO | compare_models | Command:
c:\Users\barath\AppData\Local\anaconda3\envs\dm4ml\python.exe -m papermill
C:\Users\barath\recomart-pipeline\src\05-training\comparision.ipynb C:\Users\barath\recomart-pi
pipeline\runs\prefect\20260429_164131\executed_notebooks\compare_models\comparision_executed.ipyn
b
2026-04-29 17:07:14,856 | WARNING | compare_models | STDERR:
Input Notebook: C:\Users\barath\recomart-pipeline\src\05-training\comparision.ipynb
Output Notebook: C:\Users\barath\recomart-pipeline\runs\prefect\20260429_164131\executed_notebo
oks\compare_models\comparision_executed.ipynb

Executing:    0%|          | 0/1 [00:00<?, ?cell/s]Executing notebook with kernel: dm4ml

Executing: 100%|██████████| 1/1 [00:08<00:00, 8.11s/cell]
Executing: 100%|██████████| 1/1 [00:08<00:00, 8.67s/cell]

2026-04-29 17:07:14,863 | INFO | compare_models | Completed stage: compare_models

```

9. Latest JSON Report Preview

```

{
  "run_id": "20260429T111342Z",
  "run_ts": "2026-04-29T11:13:42.777701+00:00",
  "project_root": "C:\\Users\\barath\\recomart-pipeline",
  "initial_validation": {
    "run_id": "20260429T111342Z",
    "run_ts": "2026-04-29T11:13:42.777701+00:00",
    "phase": "initial",
    "project_root": "C:\\Users\\barath\\recomart-pipeline",
    "raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\raw",
    "bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\bronze",
    "discovered_files": {
      "events_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\events.csv",
      "category_tree_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\category_tree.csv",
      "item_properties_part1_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part1.csv",
      "item_properties_part2_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part2.csv",
      "products_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\products_raw.json",
      "categories_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\categories_raw.json",
      "products_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet",
      "categories_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet"
    },
    "overall_status": "FAIL",
    "issues_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260429T111342Z.csv",
    "summary_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260429T111342Z.csv",
    "datasets": [
      {
        "phase": "initial",
        "dataset": "retailrocket_events",
        "rows": 2756101,
        "columns": 5,
        "errors": 0,
        "warnings": 2734104,
        "status": "PASS"
      }
    ],
  }
}

```

10. Conclusion

Based on the discovered project artifacts, the raw data storage layer is documented through the raw folder structure, identified raw files, related scripts or configuration files, and supporting logs or reports. This PDF provides submission-ready evidence for the Raw Data Storage deliverable.